

BSc in IT: Specialising in Data Science

IT3021: Data Warehousing
and Business Intelligence

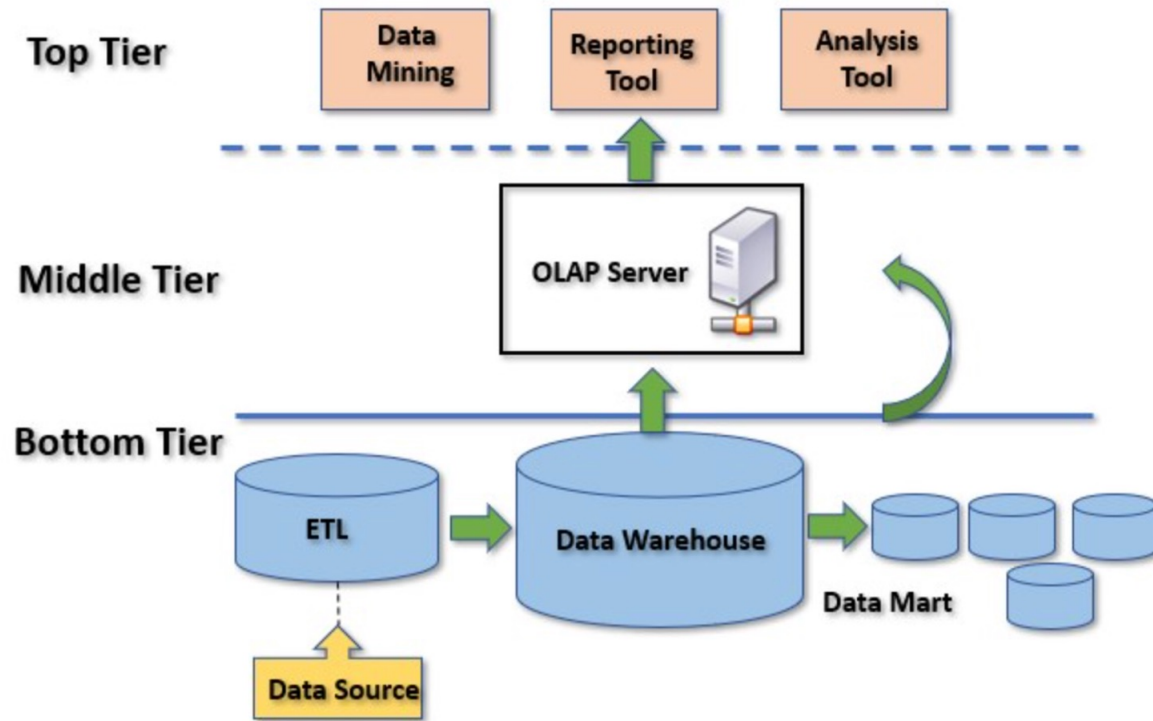
Lecture 07

OLAP Cubes

Content

- What are OLAP cubes?
- OLAP operations
- Aggregations in OLAP cubes
- Storage types
- Introduction to SSAS

Role of OLAP in Data Warehousing



- **Top-Tier:**
 - The top tier is a front-end client layer
- **Middle Tier:**
 - The middle tier is an OLAP server
 - For a user, this application tier presents an abstracted view of the database
 - Acts as a mediator between the end-user and the database
- **Bottom Tier:**
 - The database of the data warehouse servers as the bottom tier

OLAP Cubes

- A method for storing data in a multi-dimensional form
- Enables users to **analyse multidimensional data** (called facts) interactively (by browsing and querying) from **multiple perspectives** (called dimensions)
- Pre-calculates most of the queries that are typically time taking to execute over tabular databases such as aggregation, joining and grouping
 - A data structure optimized for very quick data analysis
- Specially optimized for OLAP operations:
 - Roll-up
 - Drill-down
 - Slice
 - Dice
 - Pivot

Representing Multi-dimensional Data

- Table vs. matrix

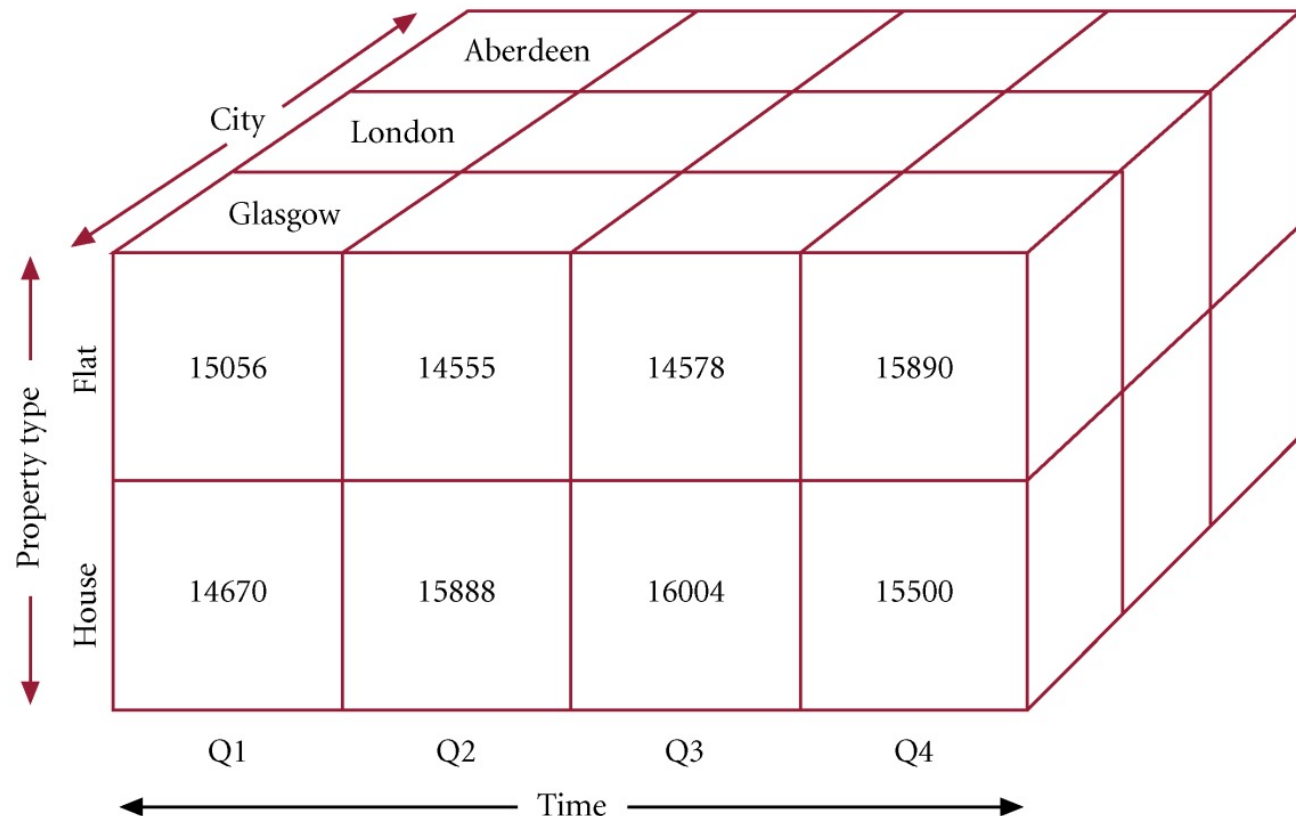
City	Time	Total Revenue
Glasgow	Q1	29726
Glasgow	Q2	30443
Glasgow	Q3	30582
Glasgow	Q4	31390
London	Q1	43555
London	Q2	48244
London	Q3	56222
London	Q4	45632
Aberdeen	Q1	53210
Aberdeen	Q2	34567
Aberdeen	Q3	45677
Aberdeen	Q4	50056
.....		
.....		

Quarter	City	Glasgow	London	Aberdeen	
	Q1	29726	43555	53210	
	Q2	30443	48244	34567	
	Q3	30582	56222	45677	
	Q4	31390	45632	50056	

Representing Multi-dimensional Data

- Table vs. cube

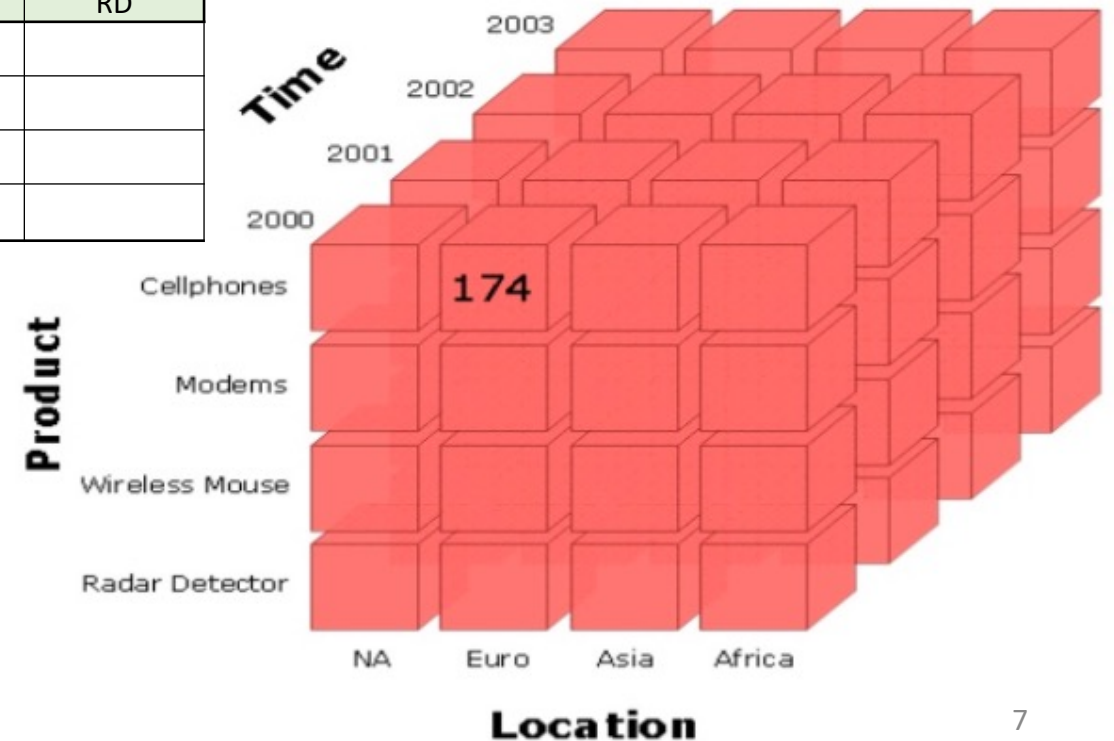
Property Type	City	Time	Total Revenue
Flat	Glasgow	Q1	15056
House	Glasgow	Q1	14670
Flat	Glasgow	Q2	14555
House	Glasgow	Q2	15888
Flat	Glasgow	Q3	14578
House	Glasgow	Q3	16004
Flat	Glasgow	Q4	15890
House	Glasgow	Q4	15500
Flat	London	Q1	19678
House	London	Q1	23877
Flat	London	Q2	19567
House	London	Q2	28677
.....			
.....			



Representing Multi-dimensional Data

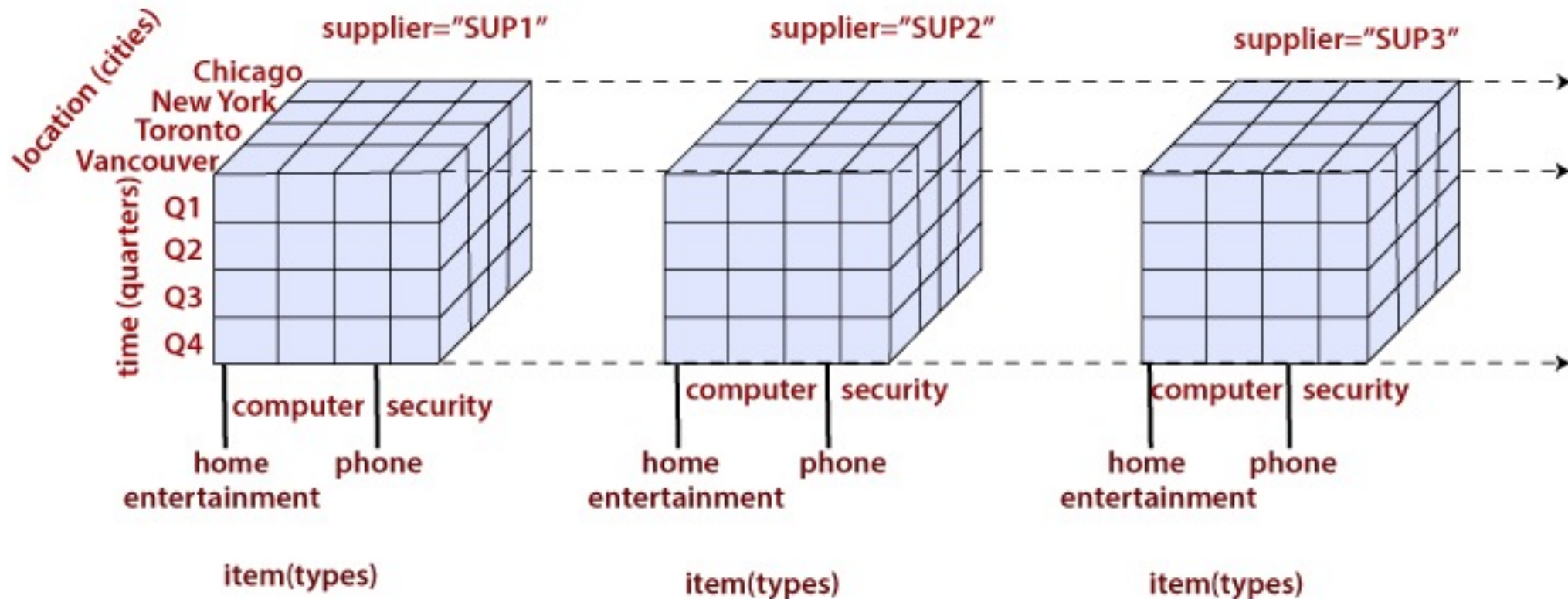
- Matrix vs. cube

Time	Location="Asia"				Location="Euro"			
	Product				Product			
	Phones	Modems	Mouse	RD	Phones	Modems	Mouse	RD
2000					174			
2001								
2002								
2003								



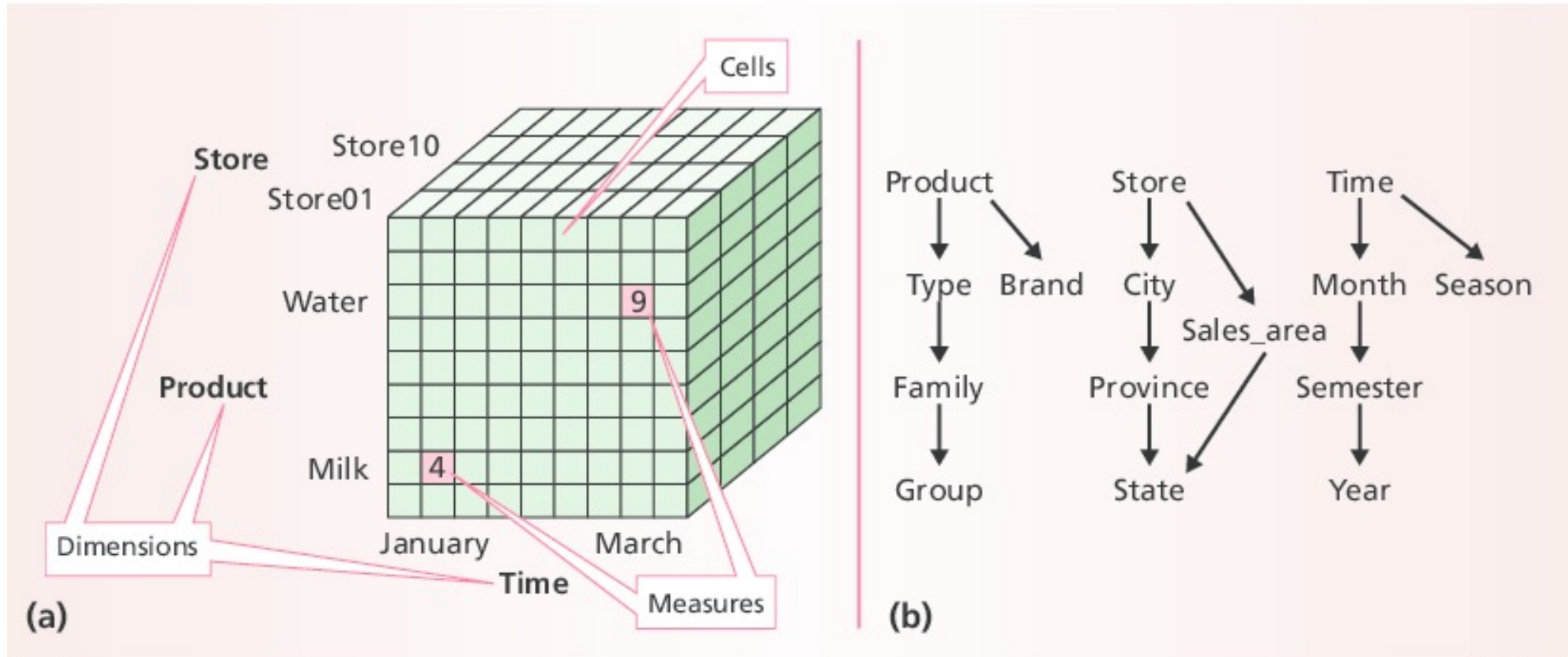
Representing Multi-dimensional Data

- Series of cubes



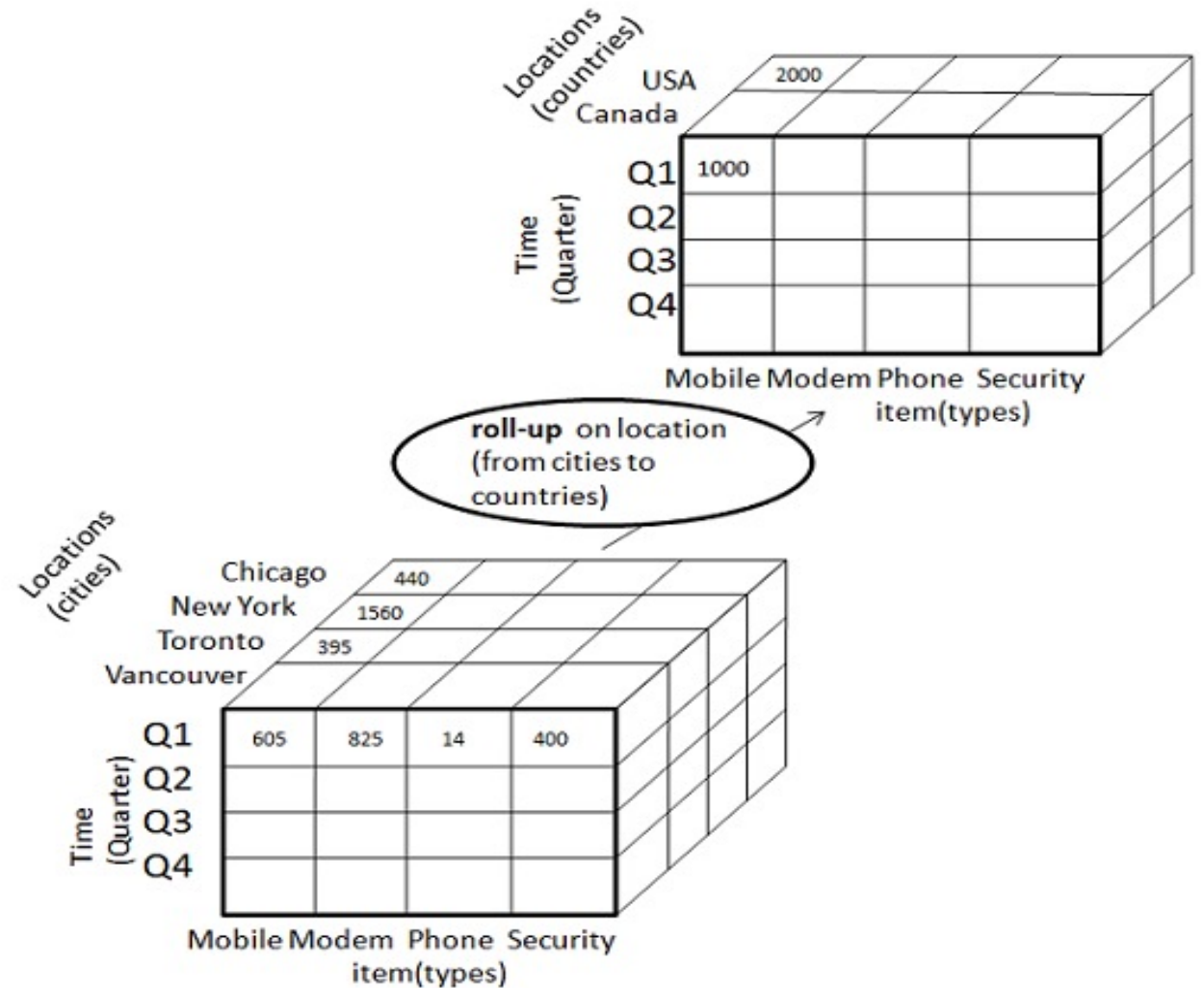
Dimensional Hierarchy

- A dimensional hierarchy defines mappings from a set of lower-level concepts to higher level concepts



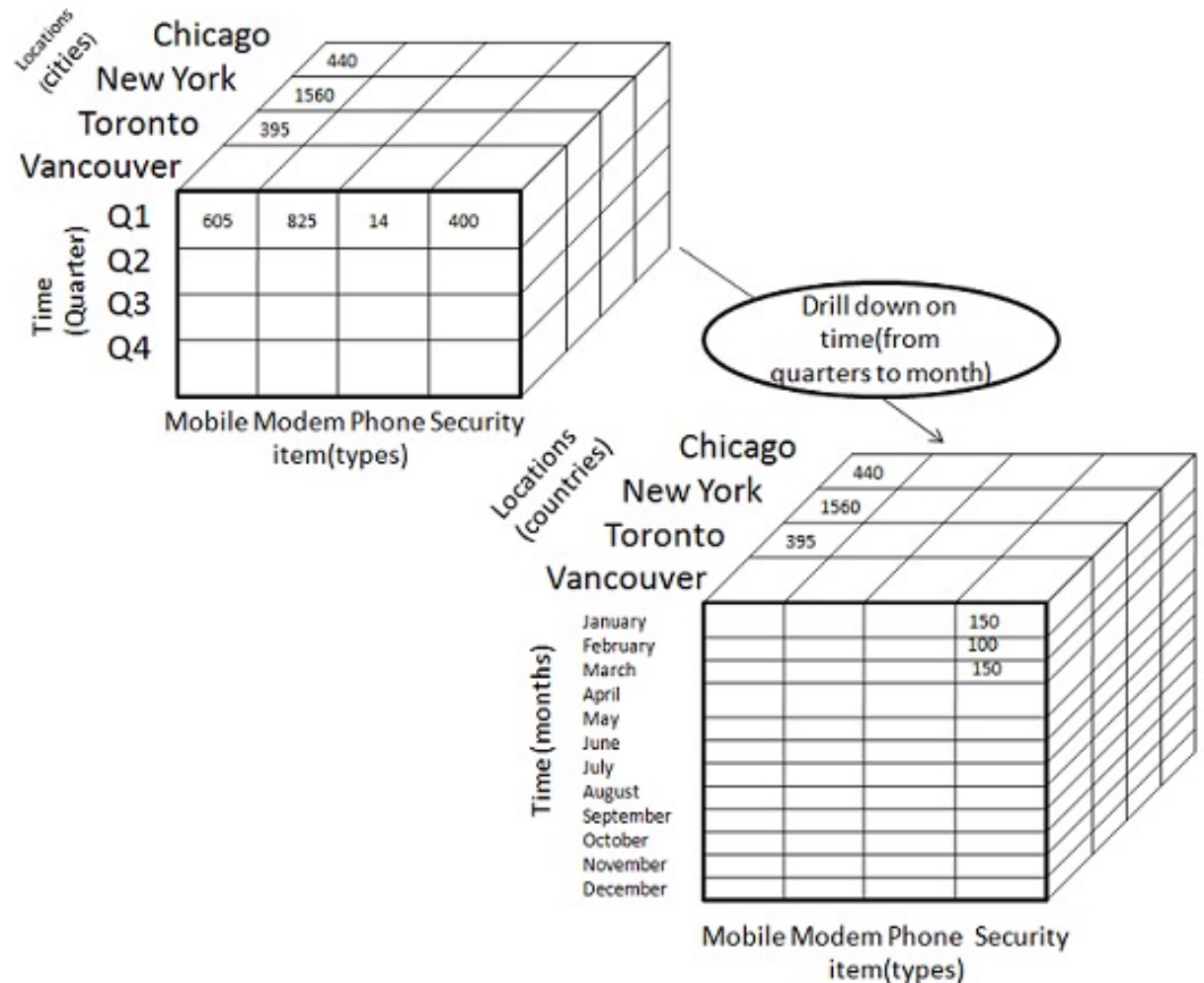
OLAP Operations: Roll-up

- Climbing up a hierarchy of a dimension to aggregate data
- Also known as consolidation or aggregation
- e.g., roll-up through country-city hierarchy in *Location* dimension



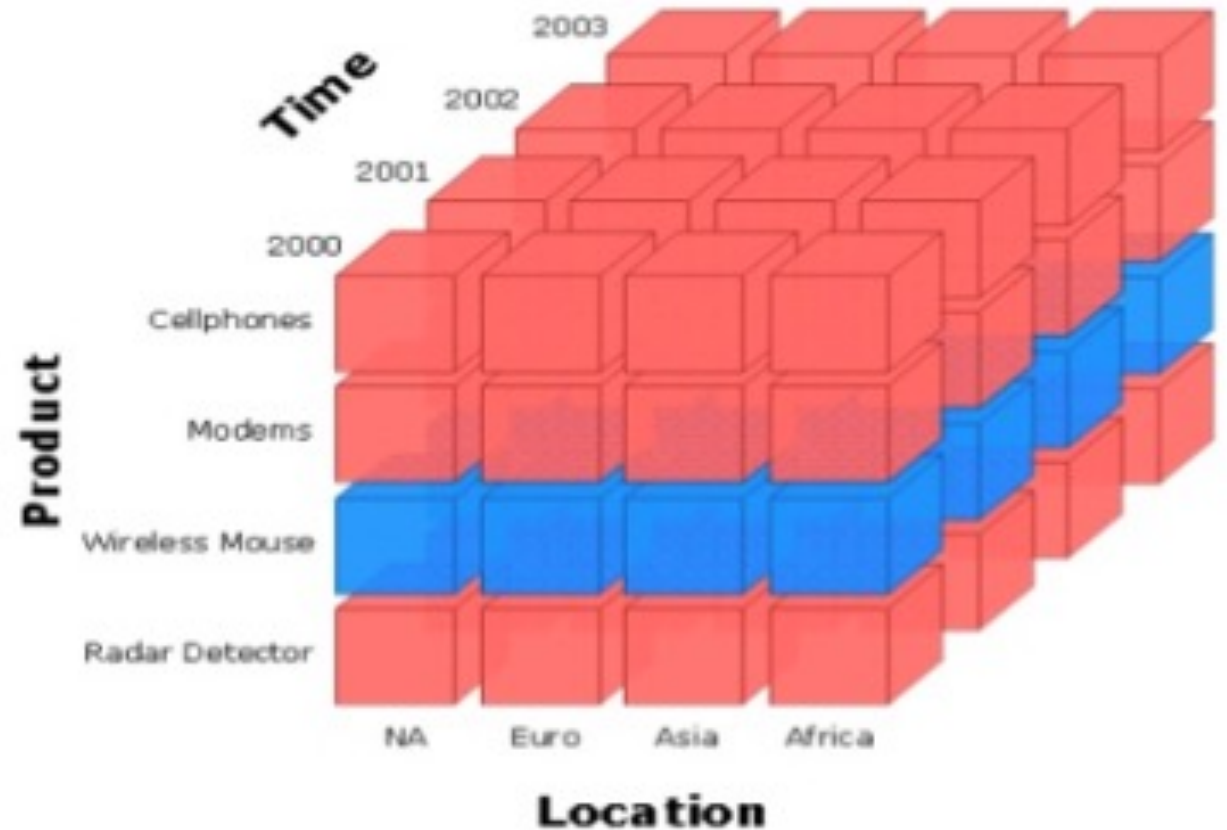
OLAP Operations: Drill-down

- Stepping down a hierarchy of a dimension allowing navigation through details
- Data is fragmented into smaller parts
- It is the opposite of the rollup process
- e.g., drill-down through quarter-month hierarchy in *Time* dimension



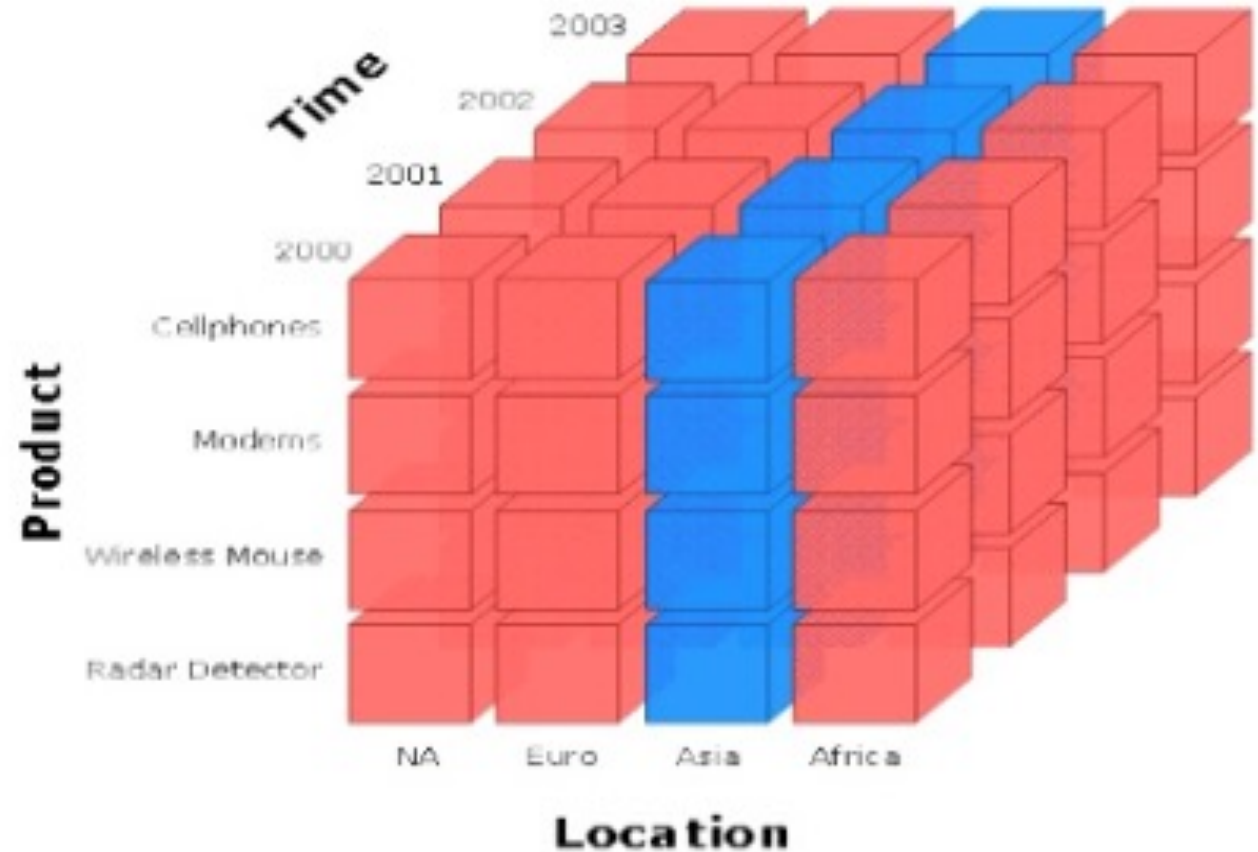
OLAP Operations: Slice

- A rectangular subset of a cube, by choosing a single value for one of its dimensions
- e.g., dimension *Product* is sliced with “Wireless Mouse” as the filter



OLAP Operations: Slice

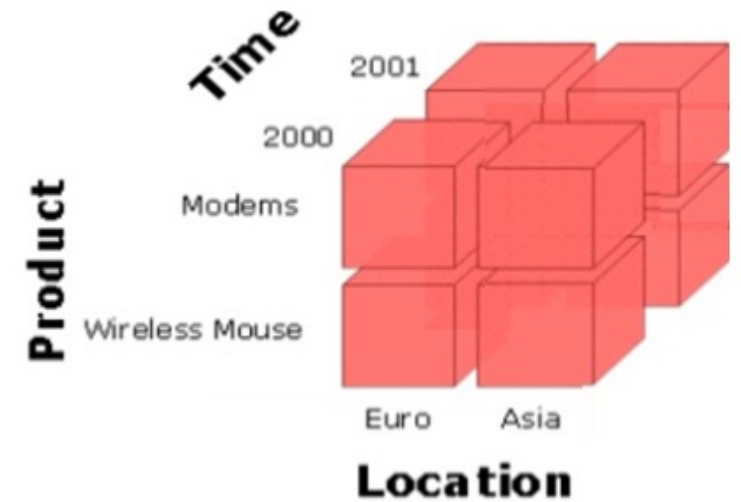
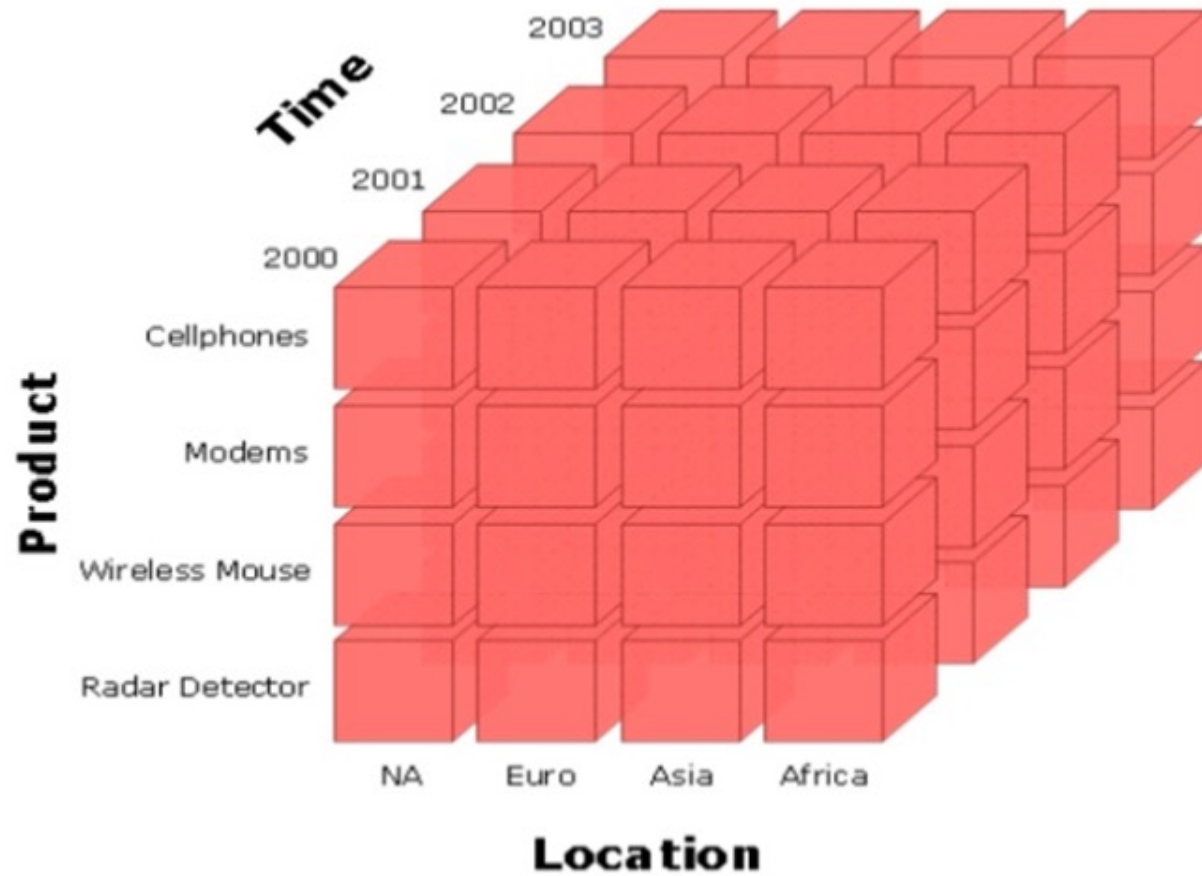
- e.g., dimension
Location is sliced with
“Asia” as the filter



OLAP Operations: Dice

- Selects two or more dimensions from a given cube and provides a new sub-cube by selecting specific values on those selected dimensions
- e.g., (Product = “Modems” or “Wireless Mouse”) and
(Time = “2000” or “2001”) and
(Location = “Euro” or “Asia”)

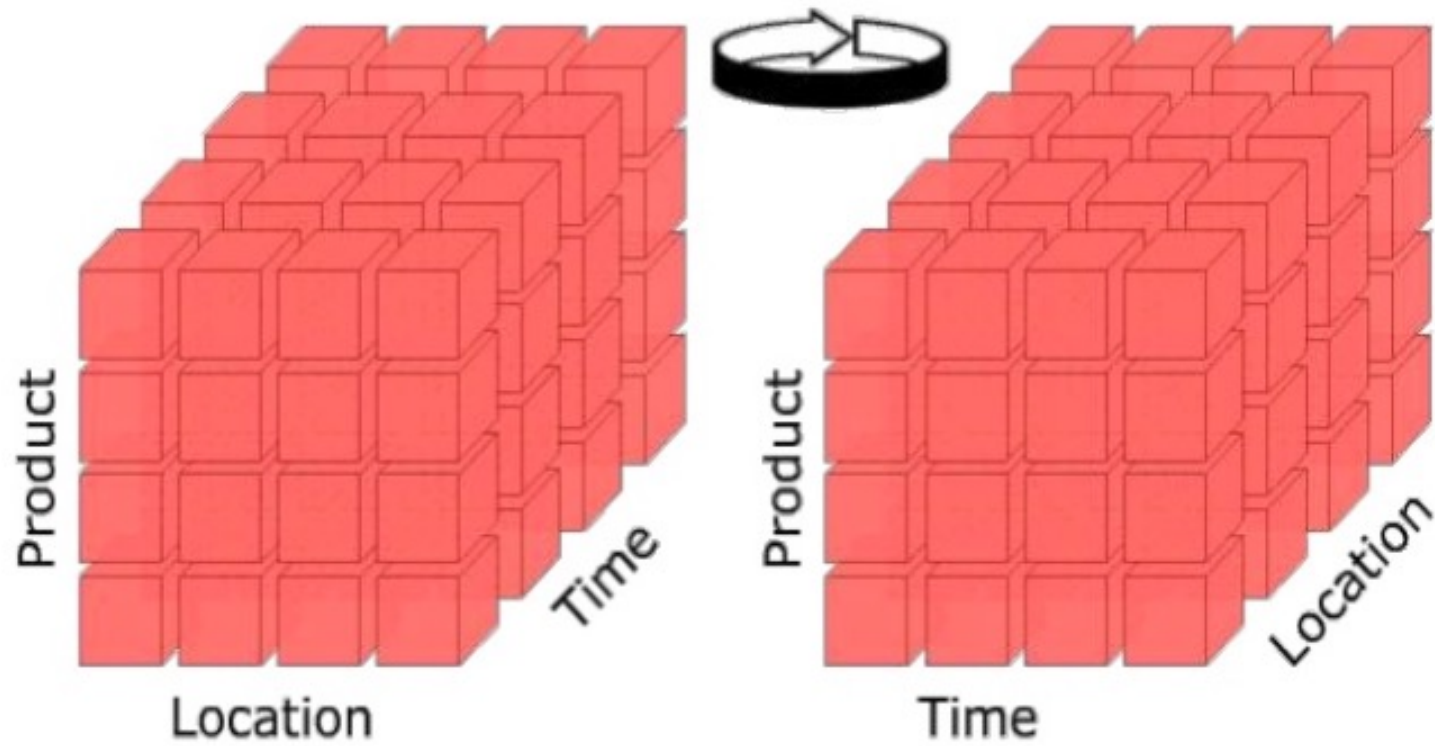
OLAP Operations: Dice



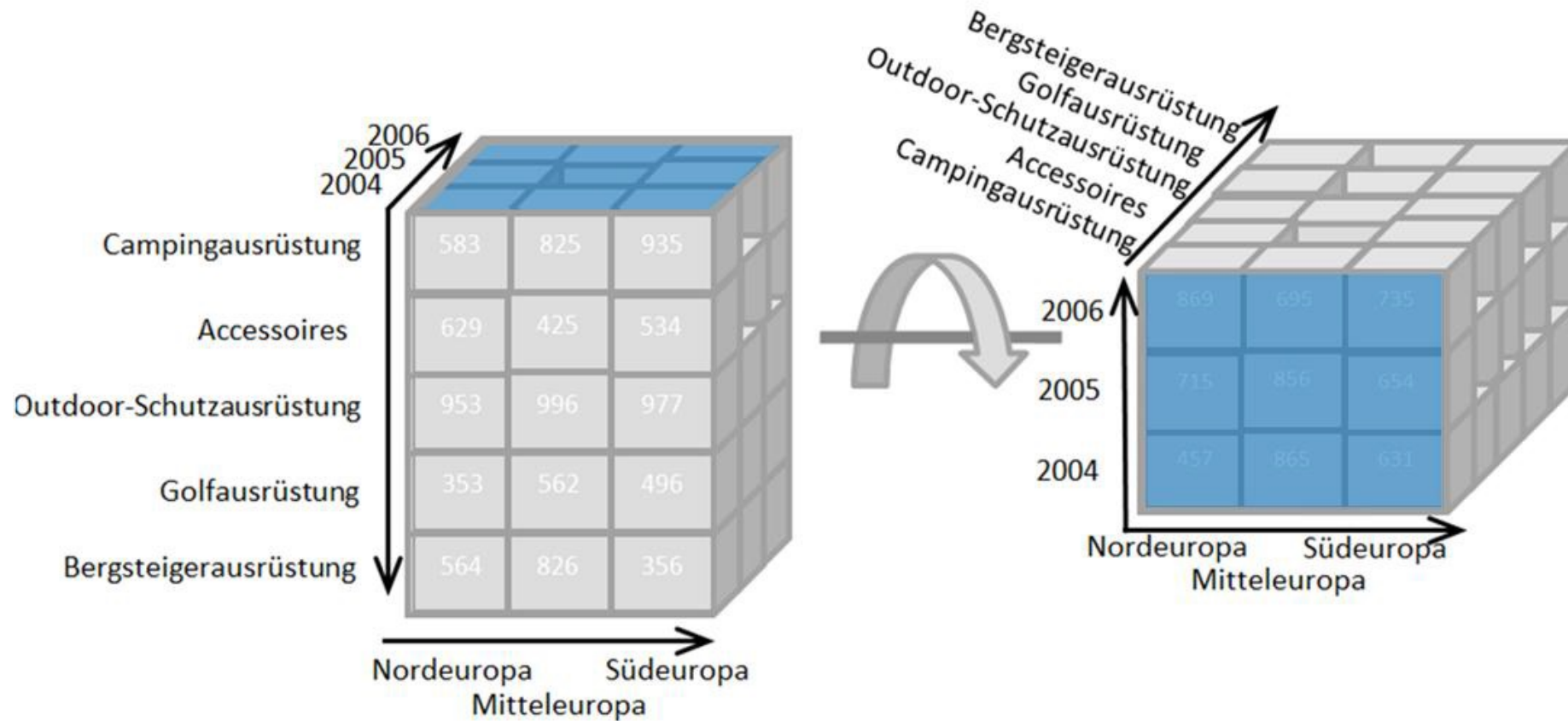
OLAP Operations: Pivot

- Pivot is a visualization operations which rotates the data axes to provide an alternative presentation of the data
 - Rotate the cube to see its various faces
- It may contain swapping the rows and columns or moving one of the row-dimensions into the column dimensions

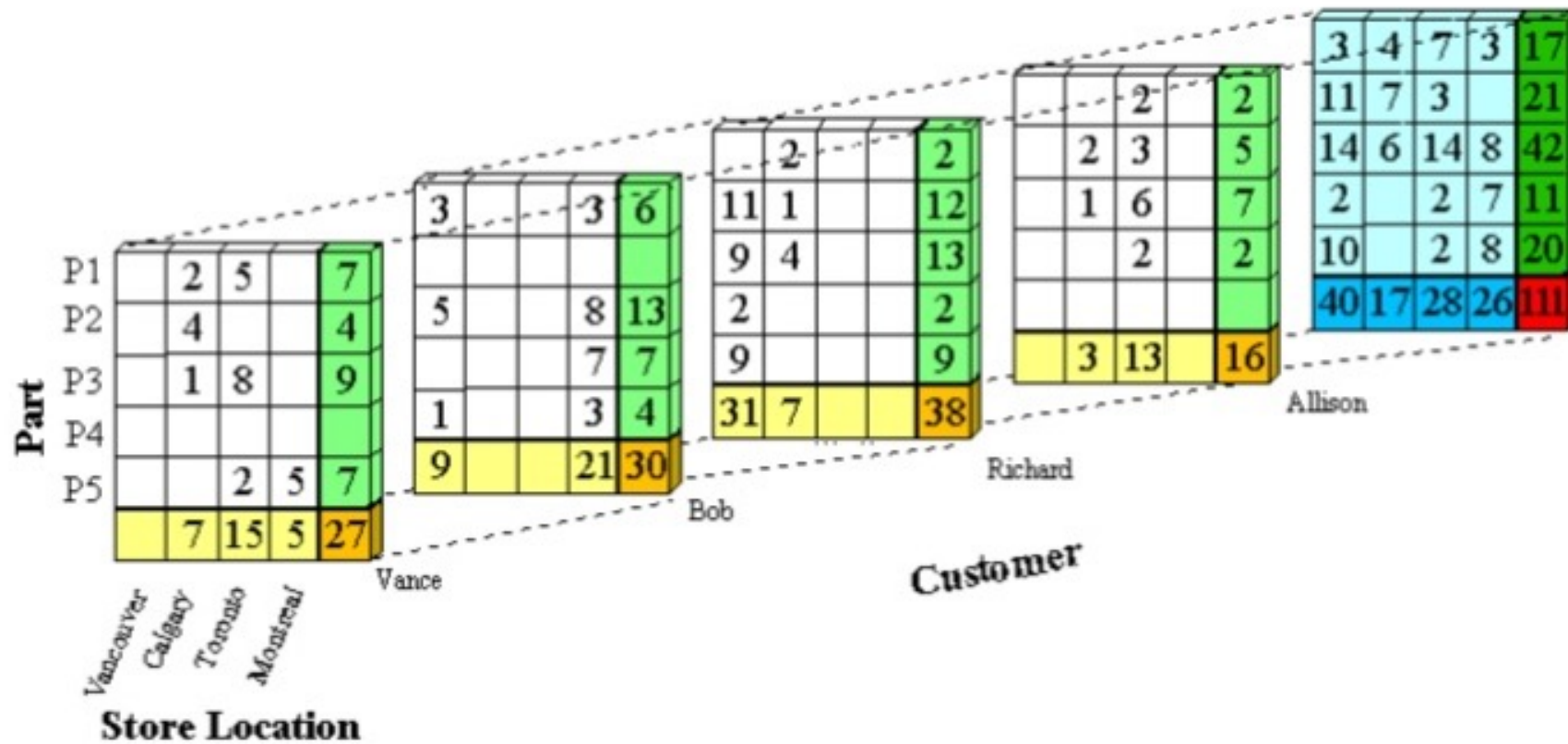
OLAP Operations: Pivot



OLAP Operations: Pivot



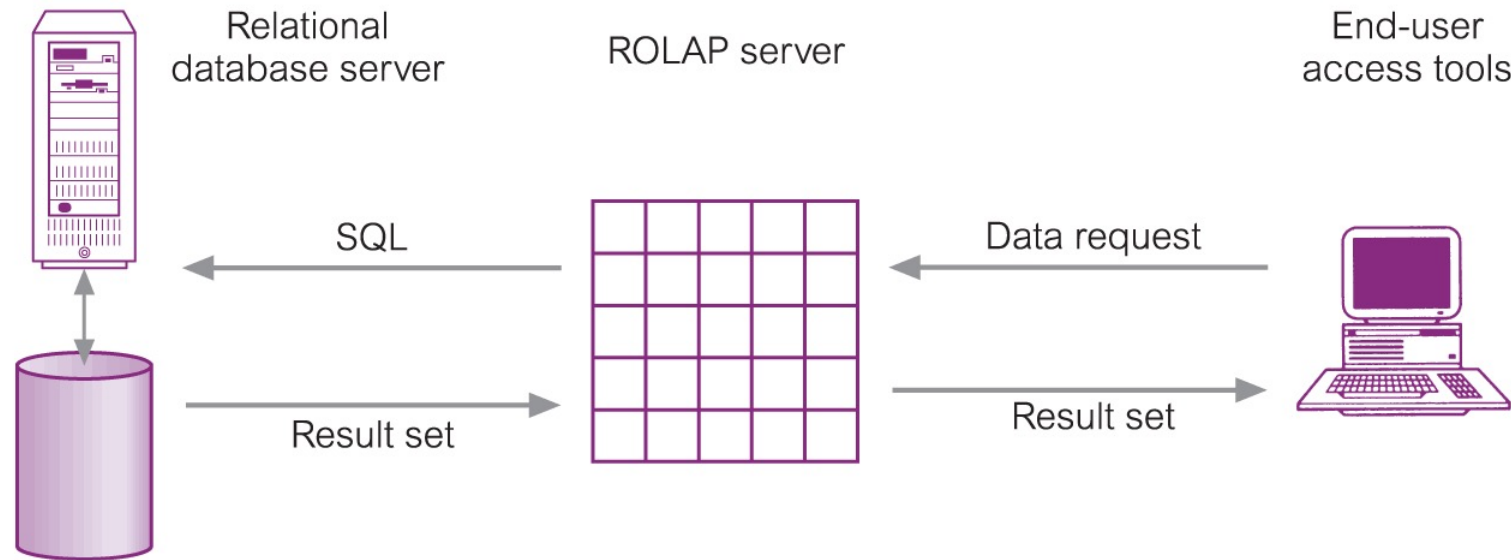
Aggregations in OLAP Cubes



OLAP Storage Types

- **ROLAP:** Relational OLAP
- **MOLAP:** Multidimensional OLAP (OLAP Cubes)
- **HOLAP:** Hybrid OLAP
- However, irrespective of the storage type, all of the above leverage same dimensional concepts

Relational OLAP

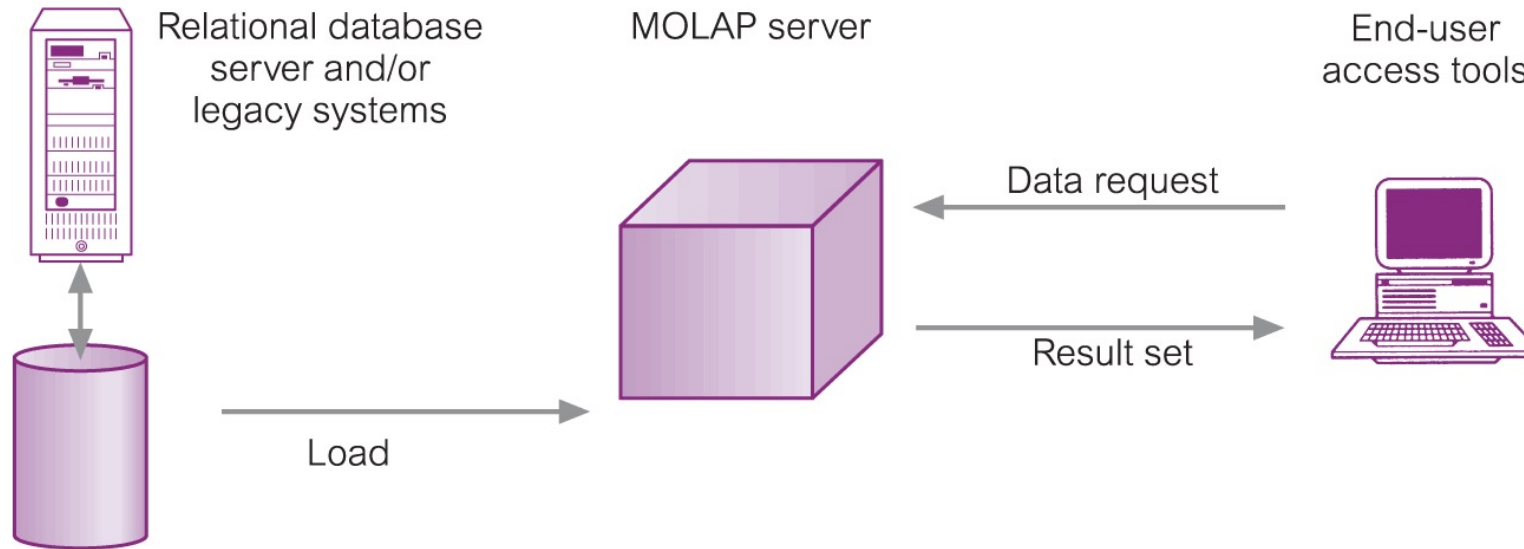


- Works primarily from the data that resides in a relational database, where the base data and dimension tables are stored as relational tables
- This model permits the multidimensional analysis of data
- Manipulates the data stored in the relational database to give the presence of traditional OLAP functionality
 - Maps functions on multidimensional data to standard relational operations
 - Data manipulation/filtering through “WHERE” clauses

Relational OLAP

- PROS:
 - Can handle large amounts of data (no limitation)
 - Leverage relational database functionalities
 - Detailed data is available
 - Highly scalable; comparatively poor query performances
- CONS:
 - Slow performance (essentially everything is a query)
 - Limited by SQL functionalities (However, vendor may provide specific out-of-the-box features)

Multidimensional OLAP

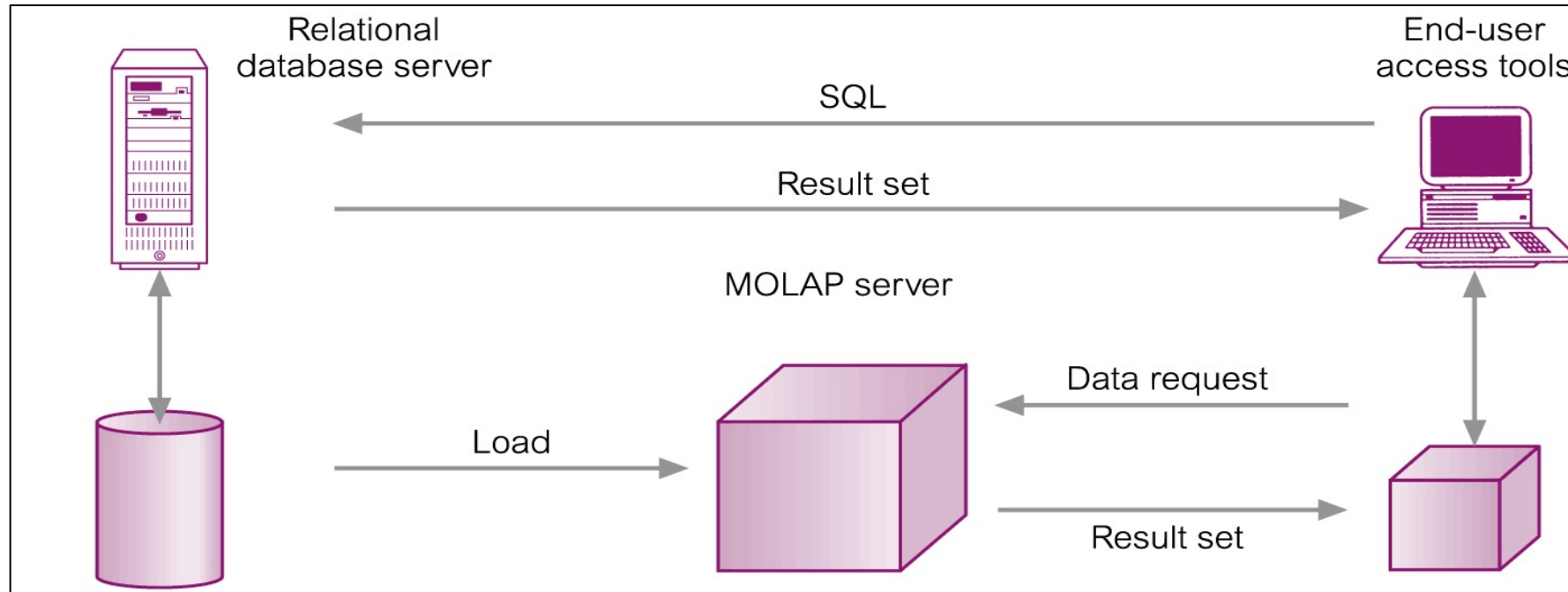


- Based on a native logical model that directly supports multidimensional data and operations
- Uses multidimensional storage engines (arrays) to store data
 - Proprietary formats
- One of the significant distinctions of MOLAP against a ROLAP is that data are summarized and are stored in an optimized format in a multidimensional cube, instead of in a relational database

Multidimensional OLAP

- PROS:
 - Excellent performance
 - Pre-calculated aggregations (fast data retrieval)
 - Can perform complex calculations and return results quickly
 - Optimal for OLAP operations (slicing, dicing, roll-up, drill-down)
- CONS:
 - Limited in amount of data (maintains a copy)
 - Additional investment (proprietary technologies)
 - Latency is high (refresh data only when processed)

Hybrid OLAP

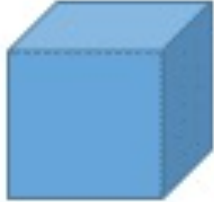
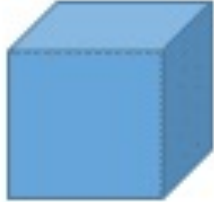
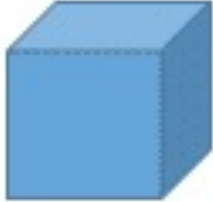
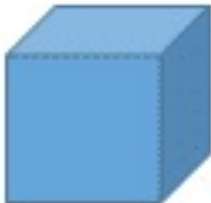
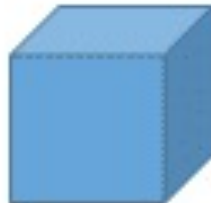
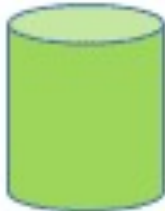
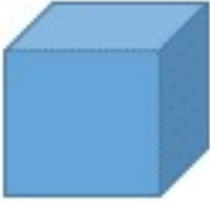


- A combination of advantages of both ROLAP and MOLAP

Hybrid OLAP

- PROS:
 - Data stored in relational format (ROLAP): minimum latency
 - Aggregations are stored in cubes: better query performance
 - It offers higher scalability of ROLAP and faster computation of MOLAP
 - HOLAP servers allows to store large data volumes of detailed information
- CONS:
 - HOLAP architecture is very complicated because it supports both MOLAP and ROLAP servers

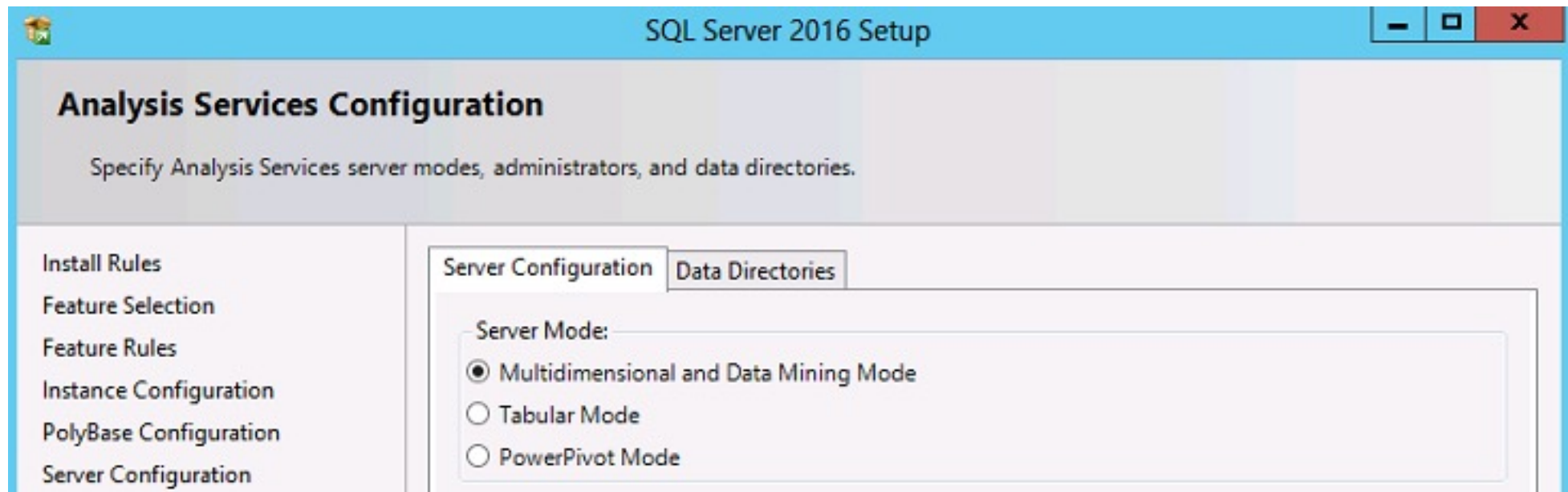
ROLAP vs. MOLAP vs. HOLAP

	MOLAP	HOLAP	ROLAP
Cube Structure			
Preprocessed Aggregates			
Detail-Level Values			

 Multidimensional Storage  Relational Storage

SQL Server Analysis Services

- 3 Server Modes



SQL Server Analysis Services

- SSAS Multi-dimensional mode has several storage modes
 - Real Time ROLAP
 - Real Time HOLAP
 - Low Latency MOLAP
 - Medium Latency MOLAP
 - Automatic MOLAP
 - Scheduled MOLAP
 - MOLAP

SQL Server Analysis Services

- Querying SSAS
 - **DAX**: Data Analysis Expressions
 - **MDX**: Multi-Dimensional Expressions
 - Retrieving data from SSAS cubes
 - **DMX**: Data Mining Extensions
 - Used for data mining structures (creating, processing, copying, browsing of mining structures and models)
 - **XMLA**: XML for Analysis
 - Commonly used in SSAS administration tasks such as backup or restore database, copy and move database or learning meta data information
 - Native SSAS Browser can also be used to analyse data

SQL Server Analysis Services

- Data mining with SSAS
 - Association rule
 - Clustering
 - Decision Trees
 - Regression
 - Naïve Bayes
 - Neural Network
 - Sequence Clustering
 - Time Series